

Far Western University

Central Department of Computer Science and Information Technology

Mahendranagar, Kanchanpur, Nepal



INTERNSHIP REPORT ON

"FULL STACK WEB DEVELOPMENT"

*"An Internship Report Submitted in partial fulfillment of the requirement for the Bachelor's
Degree in Computer Science and Information Technology (B.Sc.CSIT)"*

Under the supervision Of

Assistant Professor

Mr. Dipendra Kumar Air

Submitted By:

Bikram Dhami (Symbol NO: 8180113)

Submitted To:

Central Department of Computer Science and Information Technology Far Western University

Mahendranagar, Kanchanpur, Nepal

2nd September, 2025

STUDENT DECLARATION

I **Bikram Dhami** hereby declare that I have under done 3 months internship work at "**Global Space InfoTech Pvt. Ltd.**" during a period from 2 April 2025 to 2 July 2025 in partial fulfillment for the award of bachelor of science and technology from **Far Western University**. The work which is being presented in the internship report submitted to department of computer science and technology from **Far Western University** is an authentic record of training work.

This is an authentic piece of work and in case there is any query regarding the same, we shall be had responsible for answering any queries in this regard,

Sincerely,

Bikram Dhami

Registration No: SC-2021-1-1-0276

Date: 2 September 2025

CERTIFICATES



GLOBAL SPACE INFOTECH PVT. LTD

Godawari-1, Kailali

INTERNSHIP CERTIFICATE

To whom it may concern,

REF.NO: GSI/INT/2082/CS02

This is to certify that **Mr. Bikram Dhimi** has successfully completed his internship as a **Full Stack Developer** at **Global Space InfoTech Pvt. Ltd.** from **2nd April 2025 to 2nd July 2025**.

During his tenure with us, he actively contributed to various projects involving both **frontend and backend development**. He showcased strong technical skills, creativity, and problem-solving abilities, ensuring high-quality outputs throughout his internship.

He exhibited proficiency in the following areas:

- **Frontend:** React.js, Next.js, Bootstrap, Tailwind CSS, TypeScript
- **Backend:** Node.js, Laravel
- **Database:** MySQL, SQLite
- **Design:** UI/UX principles, responsive layouts
- **DevOps:** AWS, Docker
- **Tools:** Git, GitHub

His dedication, adaptability, and eagerness to learn were commendable. He consistently displayed a proactive attitude, handled challenges effectively, and showed a genuine passion for **Full Stack Developer**.

Given his performance, we believe that **Mr. Bikram Dhimi** has a promising future in the field of **Full Stack Development**. We extend our best wishes for his continued success in both academic and professional endeavors.

Issued on: 31 August 2025



Mr. Bhoj Raj Awasthi
Chief Technical Officer
Global Space InfoTech Pvt. Ltd.
Godawari-01, Kailali

Global Space Infotech Pvt. Ltd. Godawari – 01, Kailali Contact: +977- 9848460012

RECOMMENDATION

I hereby recommend that this internship report prepared under my supervision by **Bikram Dhami** entitled "**Full Stack Developer**" be accepted as fulfilling in partial requirements for the degree of Bachelors of Science in Computer Science and Information Technology. In my best knowledge, this is an original work in computer science by him.

Assistant Prof. Dipendra Kumar Air

Supervisor

Faculty of Science and Technology, Far western University

Kanchanpur, Nepal

EVALUATION

The undersigned certify that they have read and recommend to the Department of Computer Science for acceptance, an internship report titled "Full Stack Developer", submitted by Bikram Dhami (8180113) in partial fulfillment for the degree of Bachelor of Computer Science and Information Technology.

Evaluation Committee

Supervisor

Dipendra Kumar Air

Assistant Professor

Head of Department

Karn Dev Bhatt

Assistant Professor

External Evaluator

Internal Evaluator

Date: _____

ACKNOWLEDGEMENT

I owe a grateful appreciation to our professor who has given us chance and their valuable time. Thank you for your cooperation and guidance. I would like to express our gratitude to all those who gave us the possibility to complete this program. A huge thanks to 'Far Western University Faculty of Science and Technology'. For giving us chance to do this program. We are very much thankful to our project guide "Assistant Prof. Dipendra Kumar Air". Thank you to all the staffs of Department of Science and Technology. For their advice and suggestion and for giving chance to get practical knowledge also.

I would also like to express my special gratitude to **Global Space InfoTech Private Limited** for providing opportunity to do internship at this reputed organization with full support and cooperation. would also like to thank **Mr. Bhoj Raj Awasthi** CTO of **Global Space InfoTech Private Limited** for his mentorship.

And special thanks to our parents who inspires us in every up and down. We would like to express our gratitude to our kindhearted parents.

Sincerely,
Bikram Dhami

ABSTRACT

Developing a **complete web-based application** was the main objective of this internship. To achieve this, different programming languages and frameworks are used. The **frontend** involves technologies such as **HTML, CSS, JavaScript, React, Next.js, Bootstrap, and TailwindCSS**, while the **backend** focuses on frameworks like **Laravel (PHP)** along with other server-side technologies. Together, these tools help create powerful and interactive web applications.

Full Stack Development combines both frontend and backend processes. The **frontend (Next.js)** provides a dynamic and user-friendly interface, whereas the **backend (Laravel)** handles database management, API integration, and application logic. Modern approaches such as **MVC architecture** and **API-driven design** make the system more structured, scalable, and efficient. A full stack application is highly valuable because it can be accessed globally and provides real solutions to everyday problems.

This report, titled “**Full Stack Web Development (Next.js & Laravel)**”, highlights the knowledge and experiences gained during the internship at **Universal Informatics**. The training helped in understanding the complete development cycle, connecting frontend and backend, and solving real-world problems. These experiences have added significant value to my career as a future software developer.

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CHAPTER 1

INTRODUCTION

An internship program is a structured, short-term work experience offered by organizations to students or recent graduates, providing **hands-on exposure to real-world job environments**. It serves as a bridge between academic learning and professional practice, allowing interns to apply their theoretical knowledge in practical situations. These programs are carefully designed to help individuals develop relevant skills, gain industry insights, and explore potential career paths.

Internships can be either paid or unpaid and usually last from a few weeks to several months. During this period, organizations often provide mentorship, feedback, and performance evaluations, enabling interns to grow both personally and professionally. Beyond skill development, internships also help in building a strong professional network and may even lead to full-time job opportunities, making them an essential step toward a successful career.

1.1 BACKGROUND

The concept of internships has evolved significantly over time, originating from apprenticeship models in the medieval period, where young people would work under skilled artisans to learn trades. Over the centuries, this model transitioned into modern internships, especially in the 20th century, when formal education systems expanded, and the need for bridging academic knowledge with real-world application became apparent. Internships gained prominence in industries like medicine, law, and engineering, where practical experience is essential to developing competency. Today, they are a common part of career preparation in nearly every field, from business and technology to the arts and sciences. In the modern job market, internships have become a key strategy for students and recent graduates to gain relevant work experience, which has become increasingly important as employers often seek candidates with both educational qualifications and practical experience. With the rise of competitive industries and a globalized economy, internships provide a mutually beneficial opportunity: companies get access to fresh talent and new perspectives, while interns acquire valuable skills, experience, and sometimes a foot in the door for future employment. The growth of virtual and global internships recent years has also expanded opportunities, allowing individuals to intern for companies across different locations without the need for physical presence, making internships more accessible and diverse.

1.2 PROBLEM STATEMENT

One of the primary challenges faced by recent graduates and young job seekers is the lack of practical work experience. While academic institutions provide theoretical knowledge, they often fall short in offering real-world, hands-on experience needed to thrive in professional settings. This creates a gap between what is taught in the classroom and the actual demands of the job market. As a result, many graduates struggle to secure jobs, as employers tend to prefer candidates with prior experience, even for entry-level roles.

Internship programs bridge this gap by offering structured, hands-on training in professional environments. Through internships, participants can: Gain Practical Experience, Develop Job-Ready Skills, Build Professional Networks, Boost Employability and many more

1.3 OBJECTIVES

The primary objective of the internship program is to provide participants with practical, real-world experience in their chosen field, enhancing their professional skills and industry knowledge. Specifically, the program aims to:

- **Bridge the Gap Between Theory and Practice:** Offer opportunities for interns to apply academic knowledge to real-life business scenarios, developing a deeper understanding of their field.
- **Develop Professional Competence:** Help interns acquire industry-specific skills, improve soft skills (communication, teamwork, problem-solving), and build confidence in professional settings.
- **Enhance Employability:** Equip participants with the experience and practical knowledge needed to stand out in the competitive job market, increasing their chances of securing full-time positions.
- **Foster Professional Networking:** Provide interns with opportunities to build connections with industry professionals, mentors, and peers, which can lead to future job prospects.

1.4 APPLICATION

- **Skill Building:** Interns can apply their academic knowledge in practical settings, learning industry-specific tools, techniques, and methodologies.
- **Hands-On Experience:** Through real-world projects and tasks, interns develop essential workplace

skills such as communication, problem-solving, and teamwork.

- **Resume intern's Enhancement:** Practical experience gained through internships strengthens resume, making them more competitive in the job market.
- **Pipeline for Future Employees:** Internships serve as a testing ground for potential full-time hires, allowing organizations to assess interns' performance and fit within company culture.
- **Building Connections:** Internships offer valuable networking opportunities with industry professionals, colleagues, and mentors, which can lead to future job offers and career guidance.

1.5 GOALS

- **Enhance Skill Development:** Enable interns to acquire and develop industry-specific technical skills and soft skills such as communication, leadership, teamwork, and problem-solving.
- **Enhance Skill Development:** Enable interns to acquire and develop industry-specific technical skills and soft skills such as communication, leadership, teamwork, and problem-solving.
- **Bridge Academic and Practical Knowledge:** Help interns apply their academic knowledge in real-world business environments, fostering a deeper understanding of how theoretical concepts are implemented in practice.
- **Build Professional Networks:** Offer opportunities for interns to establish connections with professionals, peers, and mentors, creating a network that can support their future career growth.
- **Boost Employability:** Increase the intern's chances of securing full-time employment by equipping them with relevant experience, improving their resume, and preparing them for job interviews

1.6 TOOLS AND TECHNOLOGY USED

Various tools and technology were used during our internship training. Some of them are listed below:

1.6.1 React

React is a JavaScript library for building user interfaces using a **component-based** approach. It manages UI state efficiently with a **virtual DOM**, supports **unidirectional data flow**, and encourages reusable, testable components. In this project, React provides the foundation for composing views, handling events,

and managing local/UI state.

Key points: Components, props/state, hooks (e.g., useState, useEffect), virtual DOM, ecosystem-friendly.

1.6.2 Next.js

Next.js is a **full-featured React framework** that adds routing and rendering capabilities out of the box. It supports **file-based routing**, **SSR (Server-Side Rendering)**, **SSG (Static Site Generation)**, and **ISR (Incremental Static Regeneration)**, which improve performance, SEO, and scalability. Next.js also offers **API routes**, optimized asset handling, and built-in tooling

Why it's used: Faster initial page loads, better SEO, opinionated routing, and production-ready optimizations for React apps.

1.6.3 JavaScript

JavaScript is a lightweight, cross-platform and interpreted scripting language. It is well-known for the development of web pages, many non-browser environments also use it. JavaScript can be used for Client-side developments as well as Server-side developments. Features of JavaScript:

According to a recent survey conducted by Stack Overflow, JavaScript is the most popular language on earth. With advances in browser technology and JavaScript having moved into the server with Node.js and other frameworks, JavaScript is capable of so much more.

Functions in JS are objects. They may have properties and methods just like another object. They can be passed as arguments in other functions.

1.6.4 Tailwind CSS

Tailwind CSS is a **utility-first CSS framework** that provides low-level classes (e.g., p-4, text-center, bg-gray-100) to build custom designs directly in the markup. It reduces custom CSS overhead, ensures **design consistency**, and speeds up UI development. With features like **JIT compilation** and **responsive/variant utilities**, it keeps bundles lean and maintainable.

Benefits: Rapid prototyping, consistent design system, minimal context switching, easy theming and responsiveness.

1.6.5 PHP

PHP is a **server-side scripting language** widely used for web application development. It integrates well with web servers, is straightforward to deploy, and has strong support in hosting environments. PHP handles **request processing, business logic, and data access**, returning JSON responses for APIs or rendering server-side views when required.

Strengths: Mature ecosystem, large community, easy integration with databases (MySQL/PostgreSQL), broad hosting support.

1.6.6 Laravel

Laravel is a **modern PHP framework** that follows **MVC architecture** and emphasizes developer productivity and code clarity. It provides **routing, controllers, Eloquent ORM, migrations, validation, authentication/authorization, queues, events**, and robust tooling like **Artisan CLI**. Laravel's **Sanctum/Passport** enables secure API authentication for SPA/Next.js frontends.

Why it's used: Clean structure, rapid development, strong security features, maintainable codebase, first-class API building.

1.6.7 TypeScript

TypeScript is a superset of JavaScript that adds static typing and modern language features to help developers write safer, more maintainable code. Unlike plain JavaScript, TypeScript allows you to define types (like string, number, boolean, arrays, objects, and custom types) which the compiler checks before running, reducing runtime errors. It supports advanced features like interfaces, generics, enums, and type inference, making it powerful for building large-scale applications. Since TypeScript compiles down to plain JavaScript, it works anywhere JavaScript runs (browsers, Node.js, frameworks like React, Angular, or Next.js). In short, TypeScript makes JavaScript development more structured, predictable, and scalable while still giving you full access to the JavaScript ecosystem.

1.6.8 Frontend Backend Integration

- **API Design:** Laravel exposes **RESTful** or **JSON** endpoints (e.g., `/api/login`, `/api/posts`) secured via tokens (Sanctum/Passport) or sessions.
- **Data Flow:** Next.js (React) consumes these endpoints with `fetch`/Axios/React Query or `tanstack`, manages client state (or server components), and renders UI.
- **Auth & Security:** Token-based auth, CSRF protection, input validation, and role/permission checks reside in

Laravel; protected routes and UI states in Next.js

- **Performance:** SSR/SSG in Next.js improves perceived speed and SEO; Laravel handles caching (route/query/view cache) and database optimization.
- **DX/Build:** Tailwind accelerates styling; Next.js provides bundling and image optimization; Laravel's Artisan and migrations streamline backend development and deployments.

CHAPTER 2

REVIEW OF LITERATURE

Introduction

This section reviews findings and observations from previous studies on **incubation management systems, university portals, and related technologies**. The gathered information provides a strong foundation for understanding the importance of developing a digital incubation system for Far Western University (FWU). The project the **FWU Incubation Center** is designed to streamline the process of managing applications, applicants, and incubation-related activities within the university.

Literature Studies

A study entitled “**Web-Based Incubation Management System**” by R. Gupta and M. Tiwari (2019) focused on creating a digital platform for managing startups and incubation programs within academic institutions. The researchers highlighted that manual methods, such as paper-based registrations and offline communication, led to inefficiencies, data mismanagement, and delays in decision-making. Their proposed system allowed students to apply online, track the status of their applications, and enabled administrators to manage applicants effectively. The system improved transparency, reduced paperwork, and enhanced communication between incubators and participants.

Another relevant study titled “**University Startup and Incubation Portal**” by P. Sharma (2020) explored how digital platforms in higher education could foster entrepreneurship and innovation. The portal in their research facilitated the submission of startup ideas, team collaboration, mentor assignments, and performance tracking. According to the study, the platform bridged the gap between students and university incubation centers, providing students with easy access to resources, mentors, and funding opportunities. The research also emphasized the importance of **responsive design and secure authentication** for a better user experience. Similarly, in “**Digital Platforms for Entrepreneurship Support**” by K. Mishra (2021), the focus was on how online incubation portals empower students to showcase their projects and seek collaboration. The study highlighted the scalability of such systems, making them adaptable for both small universities and larger innovation hubs. It concluded that the digitalization of incubation processes not only improved efficiency but also encouraged more students to participate, as accessibility was no longer restricted by physical boundaries.

Conclusion

Based on the literature reviewed, it is evident that **web-based incubation systems** play a critical role in enhancing the

efficiency, accessibility, and effectiveness of managing incubation programs. Traditional methods, such as manual registration, paper-based record-keeping, and offline communication, often lead to delays, mismanagement, and lack of transparency.

The **FWU Incubation Center Website**, developed as part of this final year project, addresses these challenges by providing an integrated digital platform for students and administrators. It simplifies the application process, enables efficient applicant tracking, and ensures secure and organized data management. Furthermore, it creates a centralized system where innovation, entrepreneurship, and incubation activities can be effectively supported, thus contributing to the growth of entrepreneurial culture within Far Western University.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

System Analysis

System analysis is the process of understanding the existing problems, requirements, and designing a better solution. For this project, the focus is on developing a **website for the Incubation Center of Farwestern University (FWU)**. The aim of the system is to manage **news, events, applications, and applicants** in a structured and user-friendly way.

Existing System and Its Problems

Currently, the Incubation Center relies on **manual methods or simple notices** to share updates and manage applications. This creates several issues:

- Information is not properly organized.
- Applicants face difficulties in submitting and tracking applications.
- News and updates are scattered, causing poor communication with students and stakeholders.
- Manual data handling increases errors and inefficiency.
- No centralized platform for administrators to manage activities.

Expected Advantages of the Proposed System

The proposed website provides the following benefits:

- A centralized system to share **news, events, and announcements**.
- Online application submission for incubation programs.
- Organized applicant management for administrators.
- Improved communication between the center and students.
- Secure, responsive, and user-friendly design accessible across devices.

Feasibility Study

1. Technical Feasibility

The system is built using modern technologies:

- **Frontend:** Next.js, Tailwind CSS
- **Backend:** Laravel (API-based)
- **Database:** MySQL

These tools are reliable, scalable, and support the project requirements.

2. **Economic Feasibility**

Since open-source tools are used, development cost is minimal. The benefits, such as time-saving, efficiency, and improved communication, outweigh the cost.

3. **Operational Feasibility**

The system fulfills the needs of different users:

- **Admin** can post news, events, and manage applications.
- **Applicants** can apply online and upload required documents.
- **Visitors** can stay updated with incubation activities.

System Design

System Architecture

The system has three main user groups:

- **Admin:** Manages news, events, and incubation applications.
- **Applicants:** Submit and track applications online.
- **General Users:** View news, events, and updates from the incubation center.

All users interact through the web interface (Next.js frontend), which communicates with the Laravel backend and MySQL database.

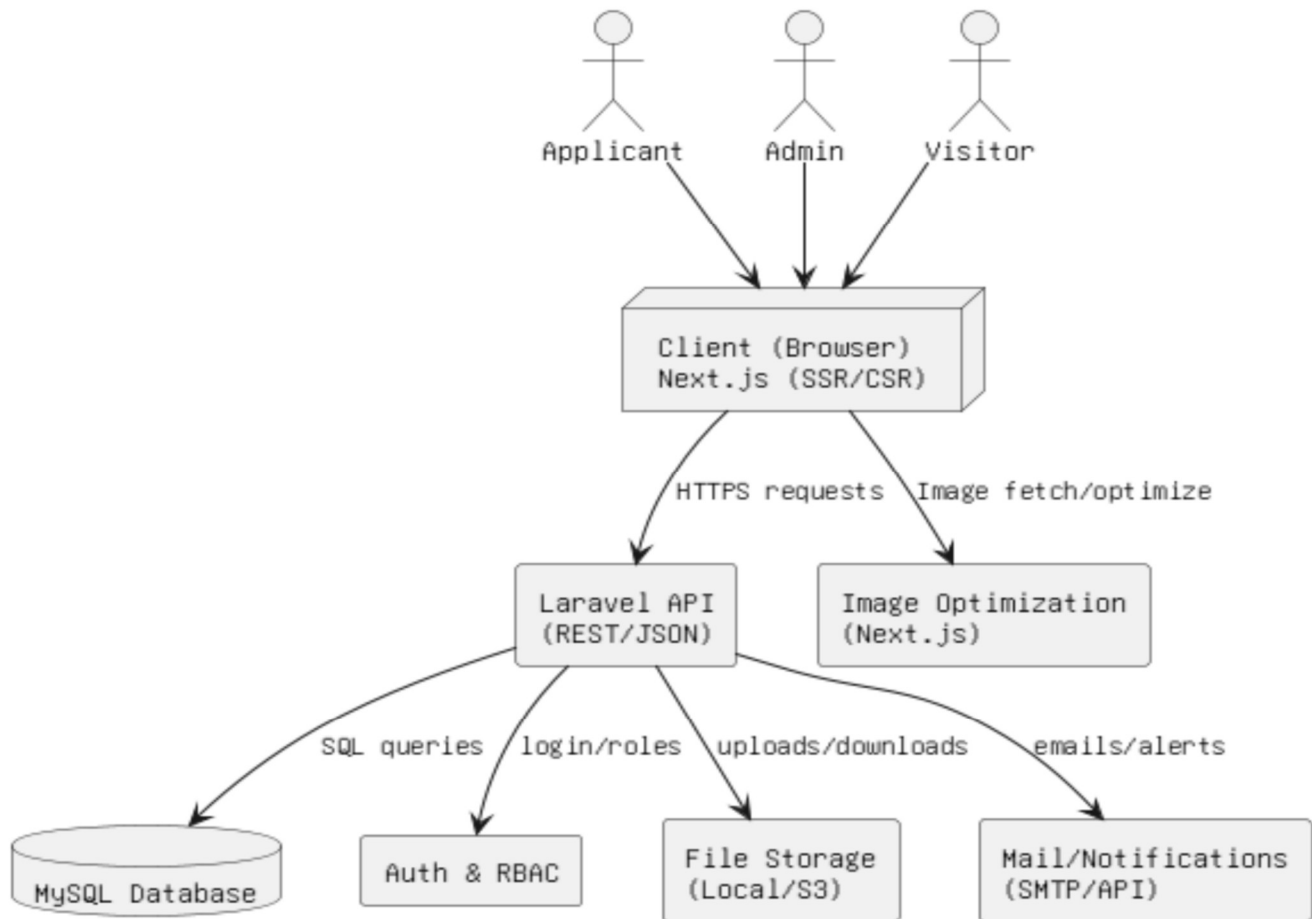


Fig:1 System Architecture

Data Flow Diagram (DFD)

1. **Level-0 DFD:** Shows interactions between users (Admin, Applicants, Visitors) and the system.
2. **Level-1 DFD:** Includes processes such as *Post News/Event*, *Submit Application*, *View Applications*, *Manage Applicants*.
3. **Level-2 DFD:** Details sub-processes such as *Approve/Reject Application*, *Upload Documents*, *Edit/Delete News*.

Database Design

The main database tables are:

1. **Admin:** Stores administrator login details.
2. **News:** Contains news posts with title, description, image, and date.
3. **Applications:** Stores program/application details (title, description, end date, status).
4. **Applicants:** Stores applicant details such as name, email, phone, and related application ID.

Entity-Relationship (E-R) Model

- An **Admin** can post many **News** and manage many **Applications**.
- An **Application** can have many **Applicants**.
- Each **Applicant** is linked to one **Application**.

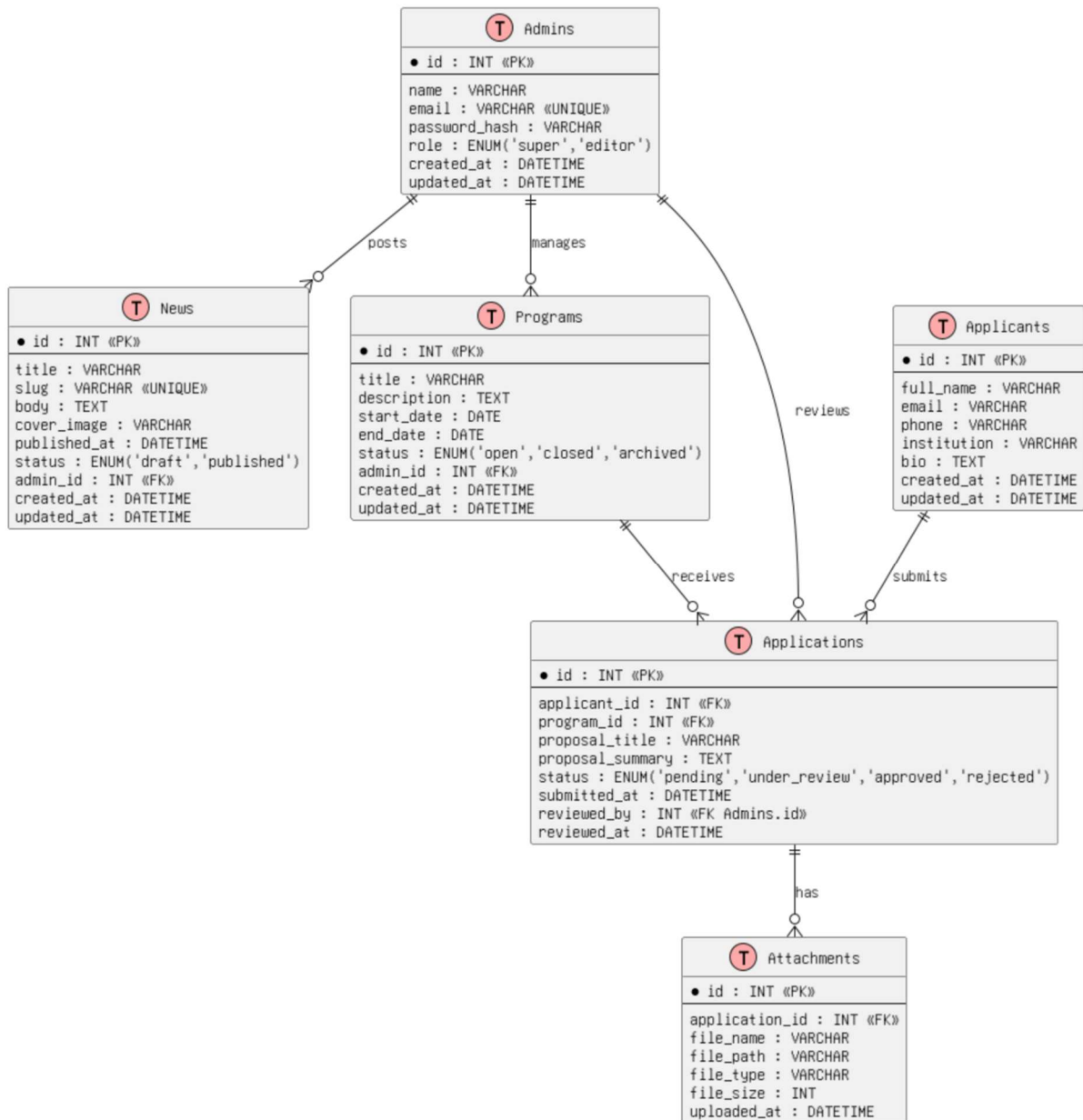


Fig2: Entity-Relationship(E-R) Model

CHAPTER 4

TRAINING WORK UNDERTAKEN

An internship program refers to the specific tasks, activities, and projects that an intern is assigned to perform during their internship. These tasks are designed to provide practical experience, develop professional skills, and enhance the intern's understanding of the industry and role. Training work can vary depending on the field of the internship and the goals of the program but generally includes the following elements:

2.1 TRAINING ATTENDED

According to Far Western University, the VIII Semester of **B.Sc.CSIT** requires students to complete an **internship program** as part of their academic curriculum. The internship must be at least **three months long** and is designed to provide students with practical knowledge and skills for building a successful career in the IT field. During the internship, students are encouraged to work on **real-world projects**, which help them apply theoretical knowledge into practice. Web development, in particular, requires the integration of multiple technologies such as **PHP, JavaScript, React, Next.js, Tailwind CSS, Laravel, and databases**, where both frontend and backend systems are combined to create complete and functional web applications.

The training that I attended In Broadway included:

- ✓ React.js
- ✓ Next.js
- ✓ Tailwind CSS
- ✓ JavaScript(ES6 +)
- ✓ TypeScript
- ✓ PHP
- ✓ Laravel Framework

2.2 INTERNSHIP DETAILS

The internship was done in **Global Space InfoTech Private Limited** under the mentorship of Bhoj Raj Awasthi Chief Technical officer.

The detail regarding internship is mentioned below:

Organization	Global Space InfoTech Private Limited
Address	Godawari-01, Kailali

Mentor	Prakash Bahadur Saud
Interned as	Full Stack Developer
Intern Duration	3 Months
Working Days	5 Days in a week

2.3 WEB DEVELOPMENT

Web development is the process of creating websites and web applications that represent an organization's brand and ensure a **user-friendly experience**. It involves both the **appearance (frontend design)** and the **functionality (backend development)** of a website. Whether developing a website, a mobile application, or maintaining web content, design and usability play a vital role. Web development can be learned to build personal projects or to pursue a professional career as a developer, creating solutions for clients. In this report, we discuss the importance of web development, its common elements, and its practical uses in modern technology.

2.4 INTERNSHIP PLACEMENT DETAILS

The internship is a mandatory requirement for the **Bachelor's Degree in Computer Science and Information Technology (B.Sc.CSIT)** under **Far Western University**. It is assigned **six credit hours** and must be completed over a minimum duration of **ten weeks (180 hours)** as part of the course curriculum.

Organizations that accept interns usually announce internship opportunities and call candidates for interviews. The selection process often includes a **casual interview, written test, and technical evaluation**. Based on performance, candidates are selected and assigned projects that allow them to apply their academic knowledge in a practical, real-world environment.

2.5 ORGANIZATION SELECTION

As per FWU (Far Western University), the internship is assigned six credit hours (minimum of ten weeks or 180 hours long). As per the organization selection process includes two major requirements which are as follows:

- Organization must be working in domain which is coherence with the final year project of the student.
- Students must be assigned minimum of 7 hours of duration per business day at the organization.

2.6 PLACEMENT

The internship candidates were placed under the supervision of the host organization, **Universal Informatics**, as assigned by the University. The supervisor for this internship program was **Mr. Bhoj Raj Awasthi** Chief

Technical Officer of the organization. The supervisor provided continuous guidance, technical support, and valuable feedback throughout the internship period. His mentorship helped the intern to conduct research, apply theoretical knowledge in practical work, and successfully fulfill the objectives of the assigned project.

2.7 ROLES AND RESPONSIBILITY

As an intern, my primary responsibilities at **Universal Informatics** were to contribute to the **design and development of web applications** as part of a **Full Stack Web Development project**. This required gaining knowledge of both **frontend technologies** such as **React.js, Next.js, Tailwind CSS, and JavaScript**, as well as **backend technologies** including **PHP and Laravel**. My role also involved understanding database management and API integration to ensure seamless communication between the frontend and backend systems.

The main responsibility was to ensure that the applications developed were **user-friendly, responsive, and aligned with client requirements**. This included designing interactive interfaces, implementing backend logic, and ensuring cross-browser compatibility. During my internship period, I worked on several practical assignments, including the development of a **College Notes Management System** and other mini-projects. These experiences enhanced my technical expertise and provided me with real-world exposure to the **complete software development life cycle (SDLC)**.

CHAPTER 5

RESULT AND DISCUSSION

3.1 DISCUSSION AND OUTCOME

During the internship period, we discussed how to analyze project requirements and implement them according to the client's needs. The projects completed during the internship were based on **modern full stack web development technologies**. The training focused on building **frontend interfaces** using React, Next.js, and Tailwind CSS, while the **backend** was developed using PHP and Laravel. This provided a complete understanding of how to design, develop, and integrate full stack applications.

Frontend Development

- **React.js:** Components, props, state management, hooks (useState, useEffect), event handling, conditional rendering.
- **Next.js:** File-based routing, server-side rendering (SSR), static site generation (SSG), API routes, optimized performance, and SEO.
- **Tailwind CSS:** Utility-first styling, responsive design, flex/grid system, typography, spacing, hover effects, and dark mode customization.
- **Frontend Integration:** Building responsive UIs, handling forms, managing authentication states, and connecting frontend with backend APIs.

Backend Development

- **PHP Basics:** Syntax, variables, operators, arrays, loops, functions, and server-side scripting.

Laravel Framework:

- MVC Architecture (Model, View, Controller).
- Routing and Middleware.
- Blade templates for rendering views.

- Database management with **Eloquent ORM** and migrations.
- Authentication and authorization.
- RESTful API development and JSON responses for frontend.
- Validation, session handling, and error management.

Project Work and Outcome

- Developed responsive web applications with modern UI using React + Next.js + Tailwind CSS.
- Built backend APIs with Laravel to manage data and authentication.
- Gained experience in frontend-backend integration using Axios/Fetch and JSON-based communication.
- Worked on real-world tasks such as form handling, database management, authentication system, and CRUD operations.
- Learned collaborative development practices like version control (Git), debugging, and testing.

3.2 PROBLEMS AND THEIR SOLUTIONS

Issues in software development often arise due to the wide variety of technologies, frameworks, and tools involved. Full Stack Development, in particular, requires a strong understanding of **frontend markups, backend methods, database structures, and API communication**. Problems can occur at any stage of development—from designing responsive interfaces to integrating server-side logic and deploying applications. It is not just about fixing technical bugs, but also about planning, communication, and meeting client expectations.

Problems

During my internship, I faced several challenges while working with **Next.js, Tailwind CSS, and Laravel**.

- In frontend development, **styling with Tailwind CSS** sometimes became confusing, as achieving pixel-perfect alignment and responsiveness required detailed attention. Designing interfaces that looked consistent across all devices was also a challenge.
- While using **React and Next.js**, I struggled with **state management, API fetching, and routing**, especially when working with **server-side rendering (SSR)** and dynamic routes.

- In backend development with **Laravel**, handling **database migrations, authentication, and API responses** was challenging at first. Errors in configuration or relationships between models often caused unexpected results.
- Another issue was **integration between frontend and backend**. Sometimes, API responses did not match the expected frontend format, which caused bugs and delays.
- Finally, working on **project deployment** and ensuring fast performance was a major challenge, as websites sometimes loaded slowly due to large assets or improper optimization.

Solutions

Through continuous learning and guidance from my supervisor, I was able to overcome these challenges.

- For **frontend styling issues**, I carefully studied Tailwind CSS utilities and used responsive design principles (flexbox, grid, breakpoints) to build layouts that worked across all devices.
- To solve **React and Next.js problems**, I practiced using hooks like `useState`, `useEffect`, and `useContext`, while also exploring Next.js features such as **API routes, SSR, and SSG**. This improved my understanding of dynamic rendering and data fetching.
- In **Laravel**, I resolved backend issues by following **MVC architecture**, learning **Eloquent ORM** for database handling, and using proper validation techniques. Debugging tools like **Laravel Telescope** and error logs also helped me identify problems quickly.
- For **frontend-backend integration**, I standardized API responses in Laravel and used **Axios/Fetch in Next.js** to ensure smooth communication. This reduced mismatches between backend data and frontend display.
- To improve **website performance**, I implemented caching strategies in Laravel, optimized images, and used Next.js optimizations such as code-splitting and lazy loading.

3.3 SDLC

SDLC is a systematic process for building software that ensures the quality and correctness of the software built. SDLC process aims to produce high-quality software that meets customer expectations. The system development should be complete in the pre-defined time frame and cost. SDLC consists of a detailed plan which explains how to plan, build, and maintain specific software. Every phase of the SDLC life Cycle has its own process and deliverables that feed into the next phase. SDLC stands for Software Development

Life Cycle and is also referred to as the Application Development life-cycle.

Here, are prime reasons why SDLC is important for developing a software system.

- It offers a basis for project planning, scheduling, and estimating
- Provides a framework for a standard set of activities and deliverables
- It is a mechanism for project tracking and control
- Increases visibility of project planning to all involved stakeholders of the development process
- Increased and enhance development Improved speed
- Improved client relations

SDLC Phases

The entire SDLC process divided into the following Phase SDLC steps:

- Phase 1: Requirement collection and analysis
- Phase 2: Feasibility study
- Phase 3: Design
- Phase 4: Coding
- Phase 5: Testing
- Phase 6: Installation/Deployment
- Phase 7: Maintenance

Here, I have explained all these Software Development Life Cycle Phases

Phase 1: Requirement collection and analysis

The requirement is the first stage in the SDLC process. It is conducted by the senior team members with inputs from all the stakeholders and domain experts in the industry. Planning for the quality assurance

requirements and recognition of the risks involved is also done at this stage.

This stage gives a clearer picture of the scope of the entire project and the anticipated issues, opportunities, and directives which triggered the project.

Requirements Gathering stage need teams to get detailed and precise requirements. This helps companies to finalize the necessary timeline to finish the work of that system.

Phase 2: Feasibility study

Once the requirement analysis phase is completed the next sdlc step is to define and document software needs. This process conducted with the help of 'Software Requirement Specification' document also known as 'SRS' document. It includes everything which should be designed and developed during the project life cycle.

Phase 3: Design

In this third phase, the system and software design documents are prepared as per the requirement specification document. This helps define overall system architecture.

This design phase serves as input for the next phase of the model.

Phase 4: Coding

Once the system design phase is over, the next phase is coding. In this phase, programming start build the entire system by writing code using the chosen language. In the coding phase, tasks are divided into units or modules and assigned to the various developers. It is the longest phase of the Software Development Life Cycle process.

Phase 5: Testing

Once the software is complete, and it is deployed in the testing environment. The testing team starts testing the functionality of the entire system. This is done to verify that the entire application works according to the customer requirement.

During this phase, QA and testing team may find some bugs/defects which they communicate to developers. The development team fixes the bug and send back to QA for a re-test. This process continues until the software is bug-free, stable, and working according to the business needs of that system.

Phase 6: Installation/Deployment

Once the software testing phase is over and no bugs or errors left in the system then the final deployment

process starts. Based on the feedback given by the project manager, the final software is released and checked for deployment issues if any.

Phase 7: Maintenance

Once the system is deployed, and customers start using the developed system, following 3 activities occur

- Bug fixing: bugs are reported because of some scenarios which are not tested at all
- Upgrade: Upgrading the applications to the newer version of the software
- Enhancement: Adding some new features into the existing software

The main focus software of this SDLC phase is to ensure that needs continue to be met and that the system continues to perform as per the specification mentioned in the first phase.

3.4 PERFORMED TASKS

During the internship, the development of the Far Western University Incubation Center Website was assigned to me. The platform was designed to manage and showcase incubation programs, news updates, and student applications. Both the frontend and backend were worked on, which provided full stack development experience. Various stages of the development lifecycle were covered, including requirement analysis, UI/UX design, development, testing, and deployment:

Tasks in FWU Incubation Website

1. Requirement Gathering and Analysis

- Coordinated with the supervisor to understand the system requirements such as news posting, application management, and applicant profiles.
- I studied similar university incubation platforms to understand common features like **program listings, announcements, and contact information**.
- Based on my research, I prepared the system workflow and designed the database schema.

2. User Interface (UI) Design and Layout

- I designed the frontend using **Next.js and Tailwind CSS**, ensuring the layout was **responsive and mobile-friendly**.
- I created pages such as **Home, News, Gallery, and About** with a clean and academic design.
- I maintained consistent branding with suitable colors, typography, and navigation structure.

3. Frontend Development

- I developed **dynamic pages and routing** in Next.js to handle different sections of the website.
- I implemented **search and filtering** features for news and Notices using React state management.
- I used **axios and tankstack** for API integration and efficient data fetching from the backend.
- I ensured a **mobile-first approach** with Tailwind CSS, making the website functional across all devices.

4. Backend Development

- I built the backend using **Laravel**, following the **MVC architecture**.
- I implemented **authentication and authorization**, allowing admins to log in and manage content securely.
- I designed and migrated the database for applications, applicants, and news modules using Eloquent ORM.
- I created **RESTful APIs** to connect the Laravel backend with the Next.js frontend

5. Testing and Debugging

- I personally tested all frontend and backend modules to ensure smooth functionality.
- I debugged **API integration issues, form validation errors, and UI alignment problems**.
- I optimized both frontend and backend for better performance and responsiveness.
- I checked for **security flaws**, including input sanitization and protection.

6. Deployment and Integration

- I deployed the Laravel backend on a local server and connected it with the Next.js frontend.
- I configured environment files for API endpoints and database connections.

3.5 RESULT

Following are the screenshots of the result of the project done during the internship period.

Home Page

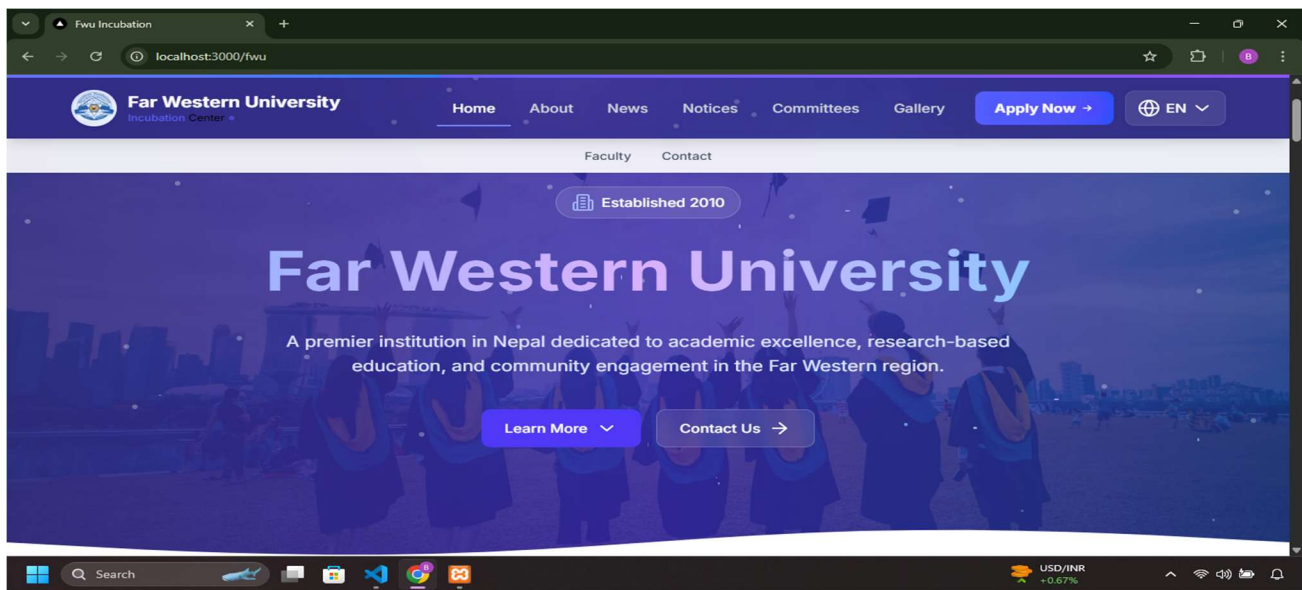


Fig.1: Homepage of website

News Page

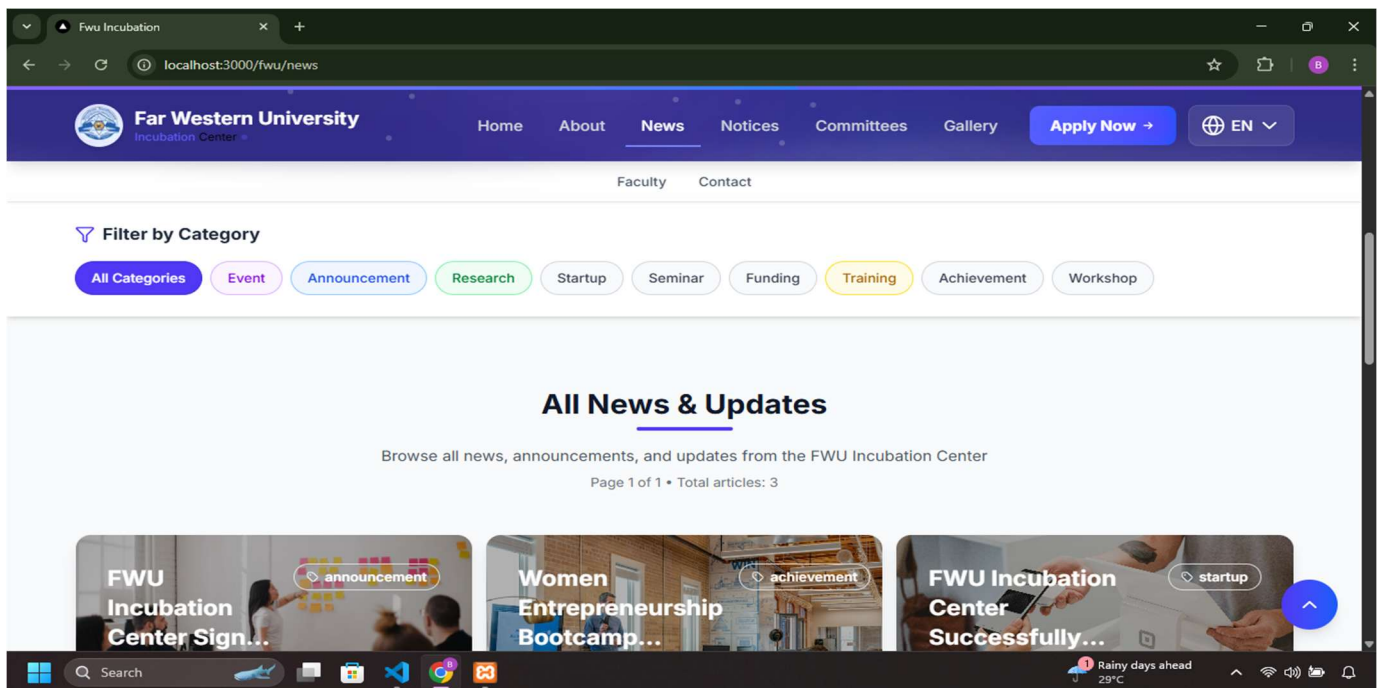


Fig 2: Nws page

Application Form Page

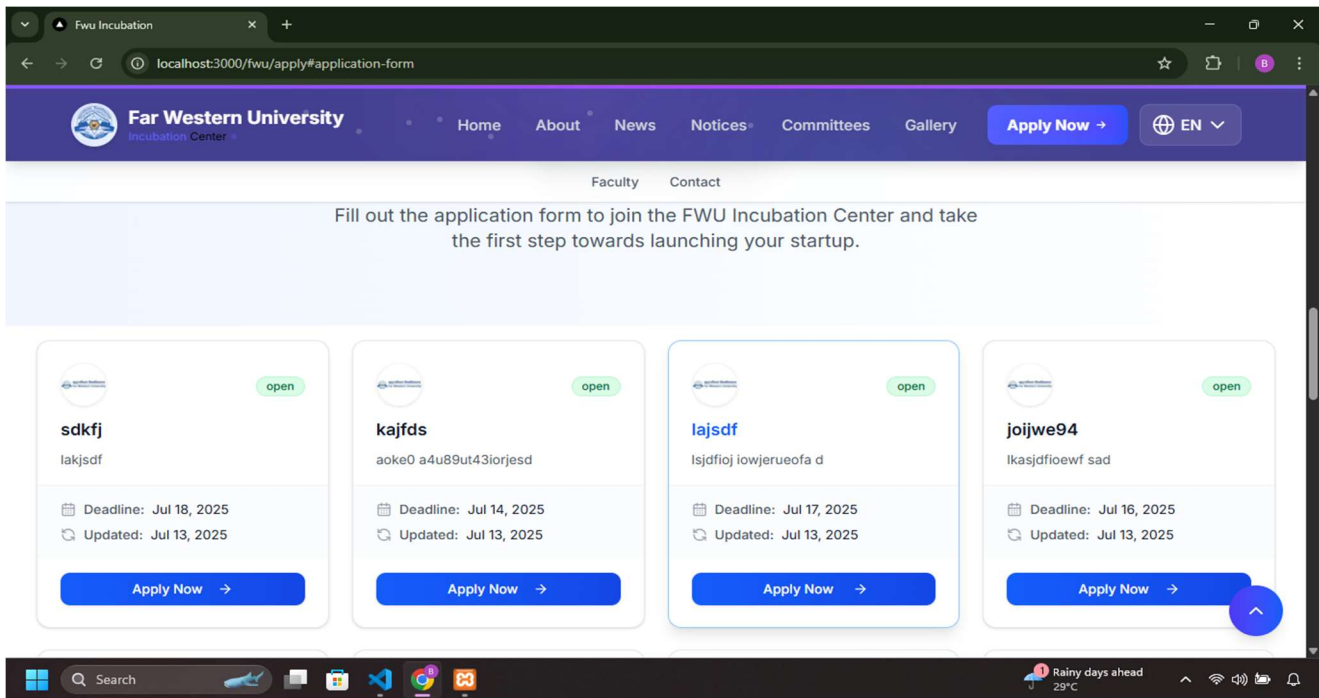


Fig 3: Application Form page

Log In Page

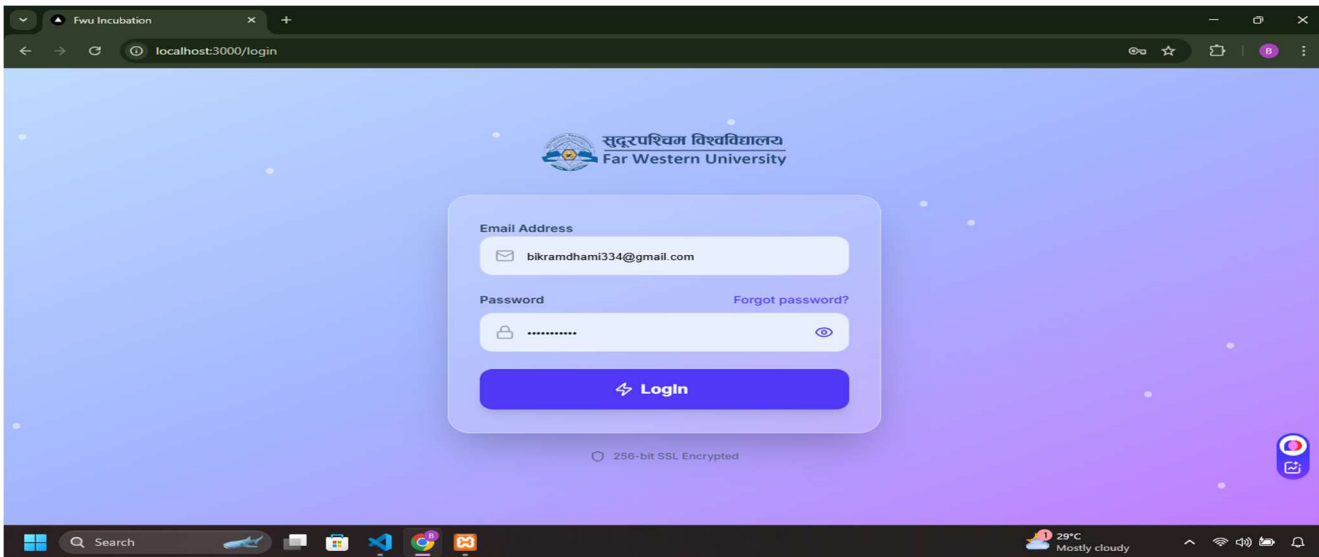


Fig 4: Log in page

Dashboard Page

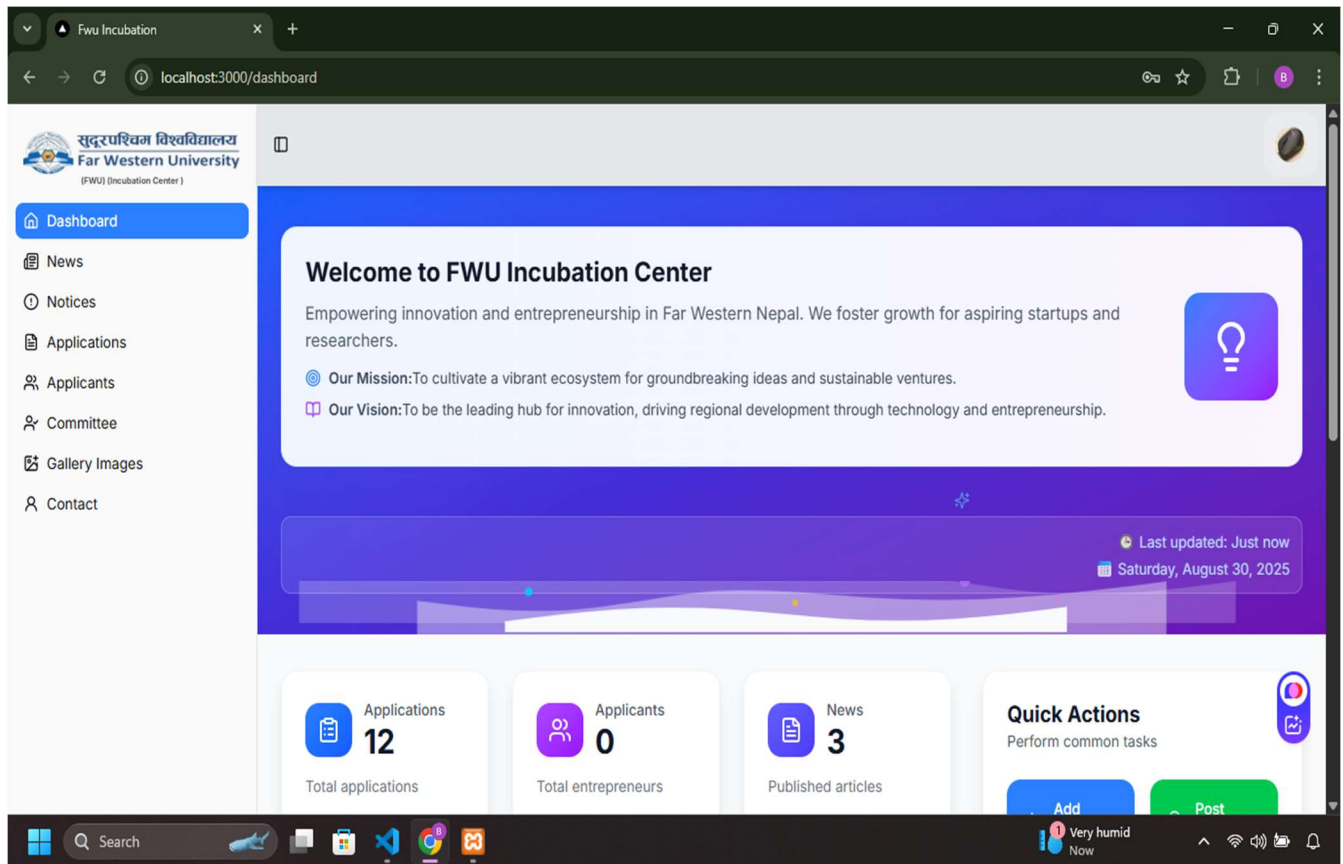


Fig 5: Dashboard Page

Profile Page

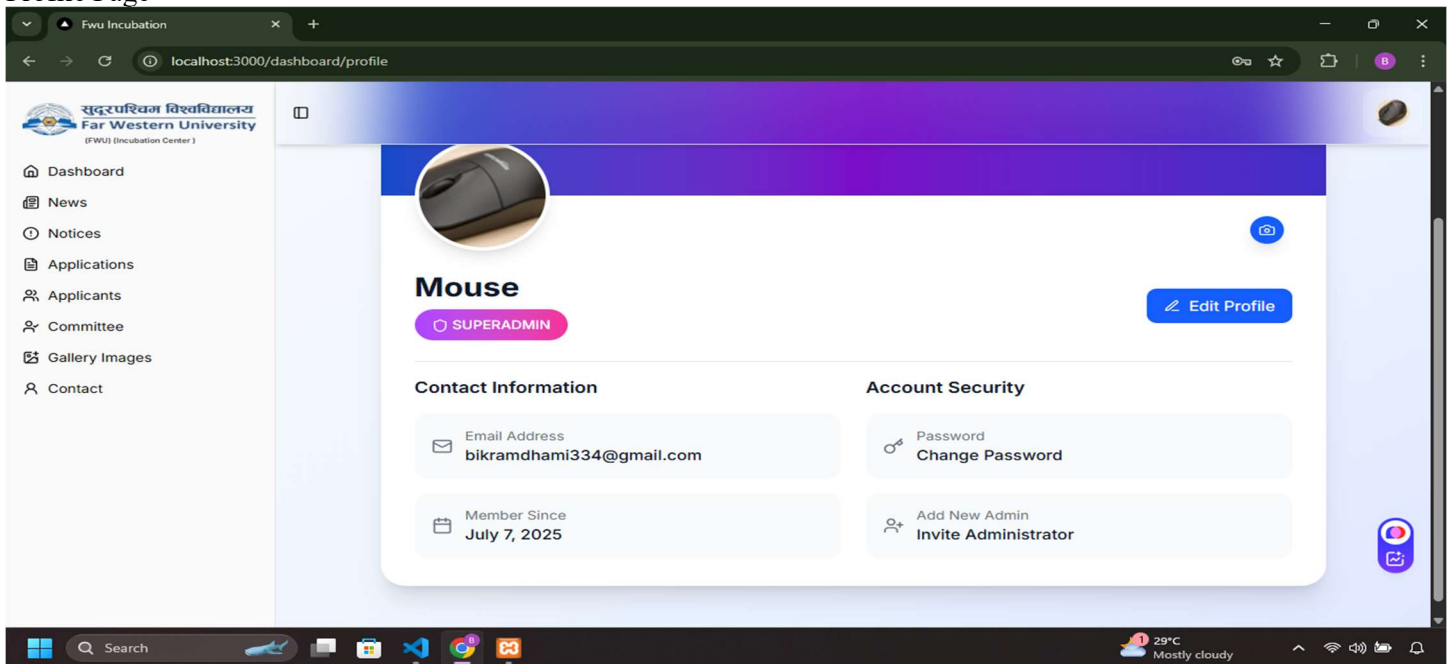


Fig 5: Profile Page

3.5 CODE SCREENSHOT

Home page

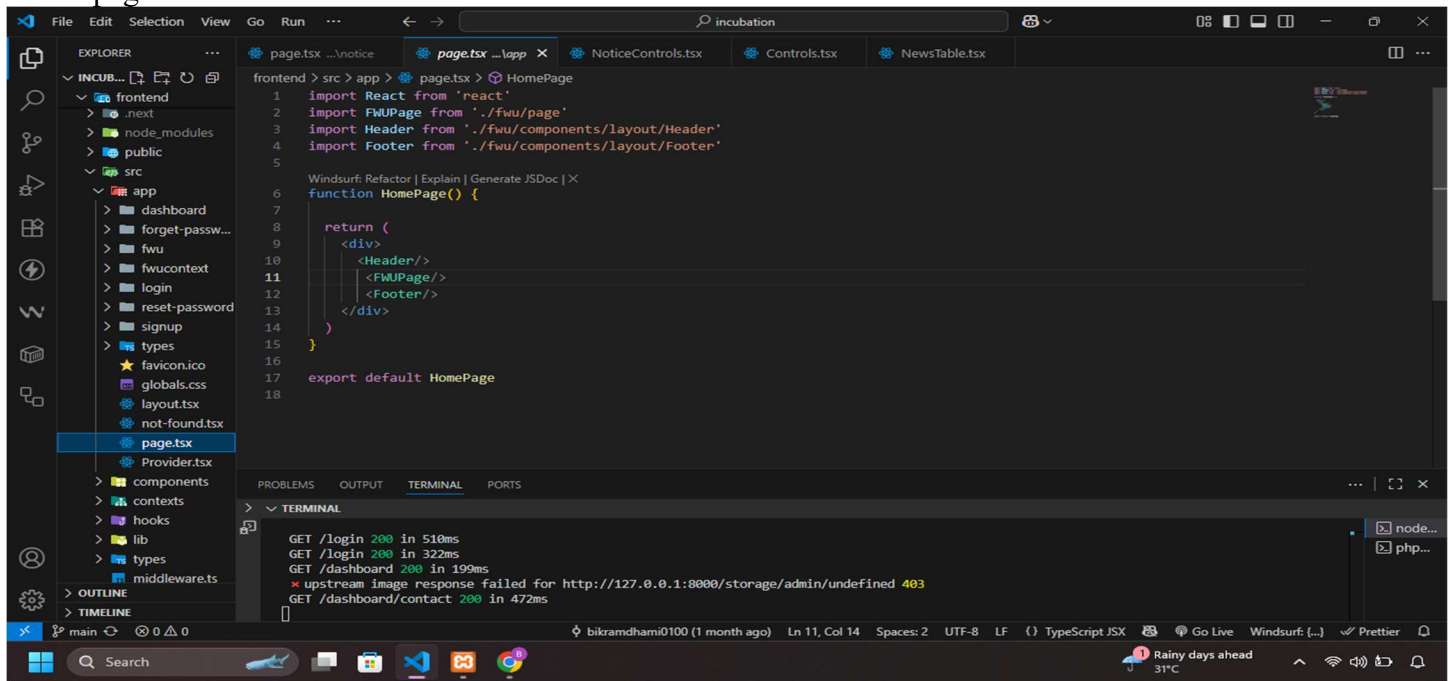


Fig:1

Dashboard Page

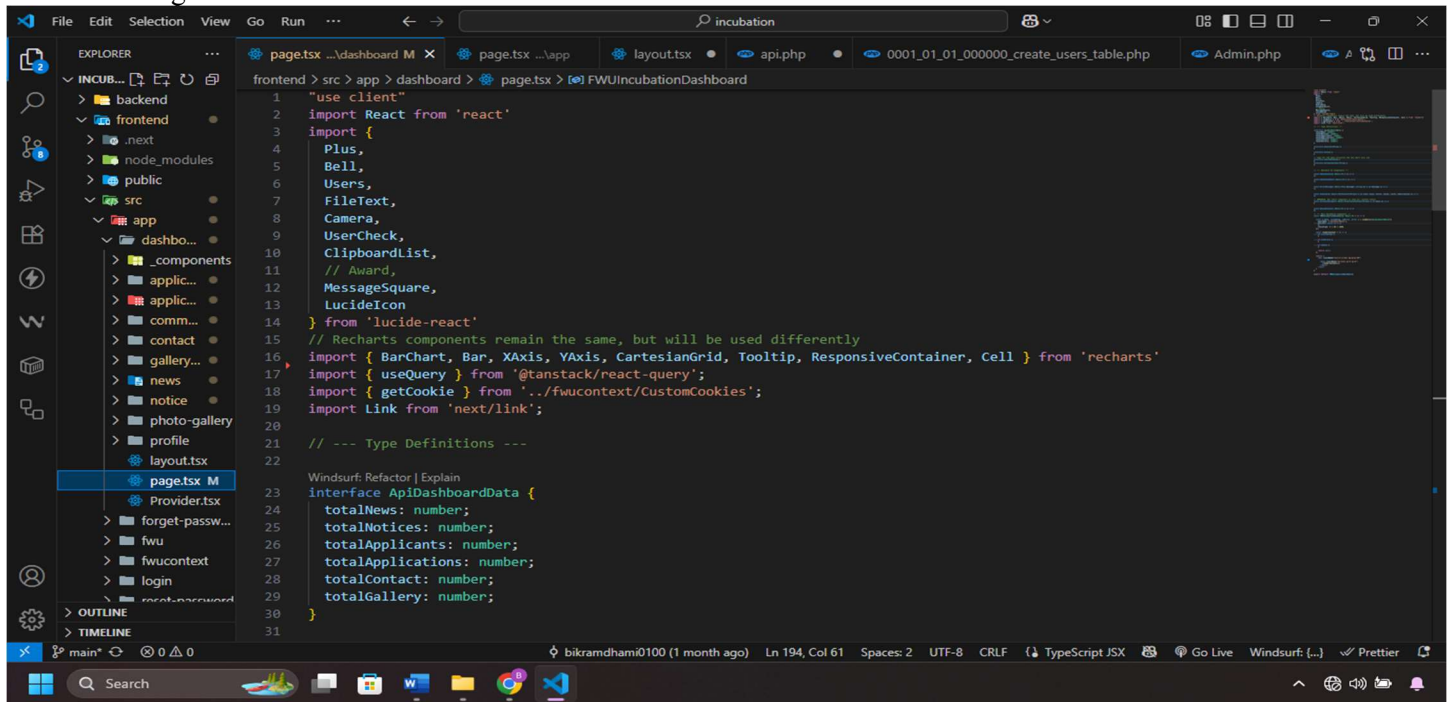
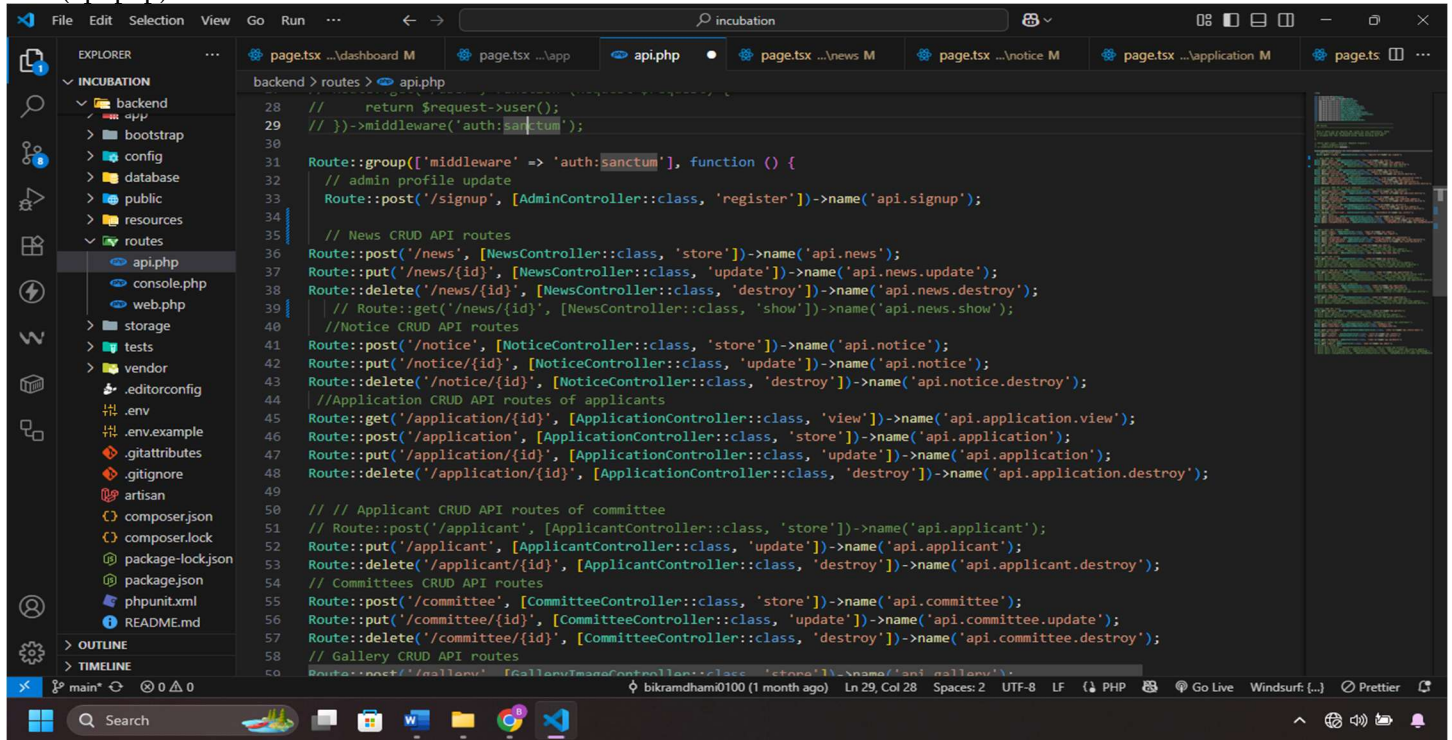


Fig:2

API (api.php)

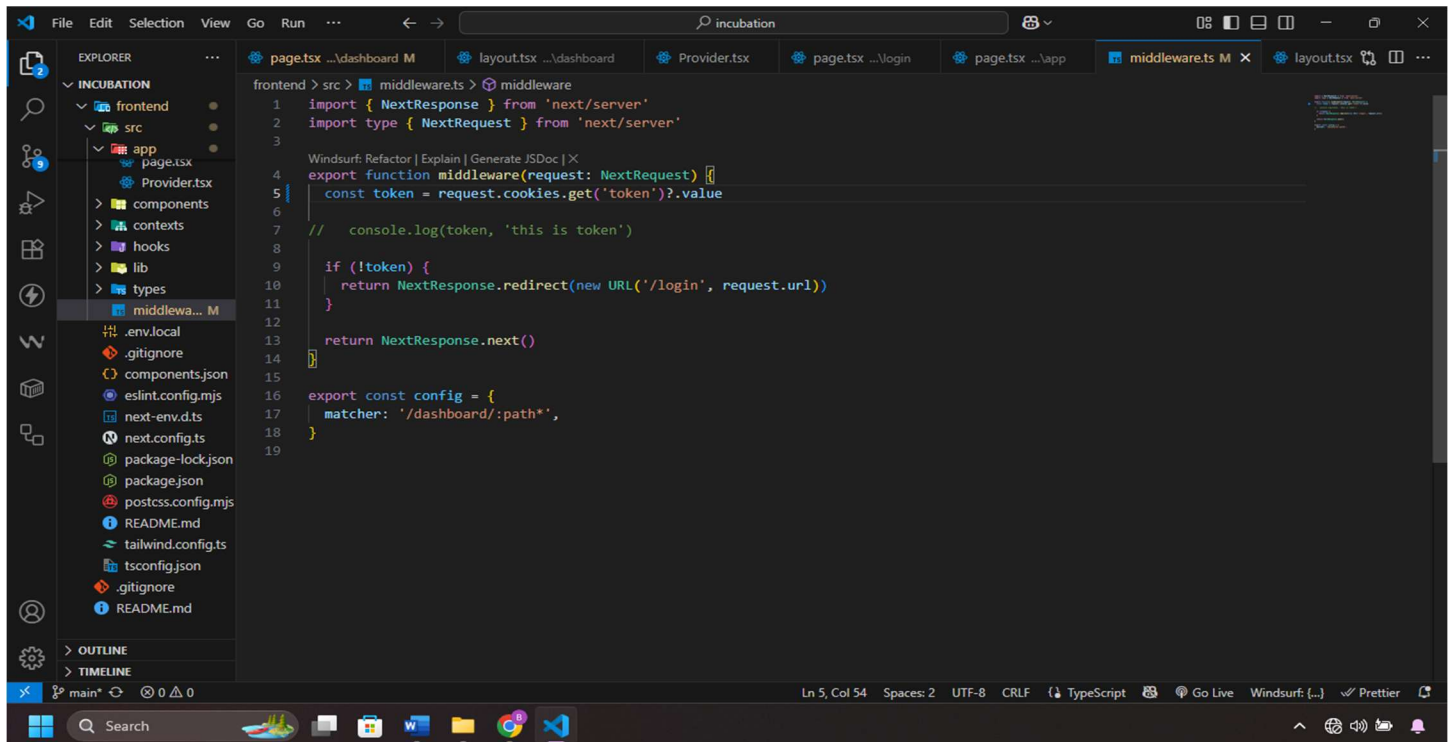


The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor window shows the `api.php` file, which contains a list of API routes. The routes are organized into groups and include comments for each endpoint. The routes are as follows:

```
28 // return $request->user();
29 // }->middleware('auth:sanctum');
30
31 Route::group(['middleware' => 'auth:sanctum'], function () {
32 // admin profile update
33 Route::post('/signup', [AdminController::class, 'register'])->name('api.signup');
34
35 // News CRUD API routes
36 Route::post('/news', [NewsController::class, 'store'])->name('api.news');
37 Route::put('/news/{id}', [NewsController::class, 'update'])->name('api.news.update');
38 Route::delete('/news/{id}', [NewsController::class, 'destroy'])->name('api.news.destroy');
39 // Route::get('/news/{id}', [NewsController::class, 'show'])->name('api.news.show');
40 // Notice CRUD API routes
41 Route::post('/notice', [NoticeController::class, 'store'])->name('api.notice');
42 Route::put('/notice/{id}', [NoticeController::class, 'update'])->name('api.notice.update');
43 Route::delete('/notice/{id}', [NoticeController::class, 'destroy'])->name('api.notice.destroy');
44 // Application CRUD API routes of applicants
45 Route::get('/application/{id}', [ApplicationController::class, 'view'])->name('api.application.view');
46 Route::post('/application', [ApplicationController::class, 'store'])->name('api.application');
47 Route::put('/application/{id}', [ApplicationController::class, 'update'])->name('api.application.update');
48 Route::delete('/application/{id}', [ApplicationController::class, 'destroy'])->name('api.application.destroy');
49
50 // // Applicant CRUD API routes of committee
51 // Route::post('/applicant', [ApplicantController::class, 'store'])->name('api.applicant');
52 Route::put('/applicant', [ApplicantController::class, 'update'])->name('api.applicant.update');
53 Route::delete('/applicant/{id}', [ApplicantController::class, 'destroy'])->name('api.applicant.destroy');
54 // Committees CRUD API routes
55 Route::post('/committee', [CommitteeController::class, 'store'])->name('api.committee');
56 Route::put('/committee/{id}', [CommitteeController::class, 'update'])->name('api.committee.update');
57 Route::delete('/committee/{id}', [CommitteeController::class, 'destroy'])->name('api.committee.destroy');
58 // Gallery CRUD API routes
59 // Route::post('/gallery', [GalleryImageController::class, 'store'])->name('api.gallery');
```

Fig:3

Middleware



The screenshot shows the VS Code editor with the file explorer on the left displaying the project structure. The main editor window shows the `middleware.ts` file, which contains a Next.js middleware function. The code is as follows:

```
1 import { NextResponse } from 'next/server'
2 import type { NextRequest } from 'next/server'
3
4 export function middleware(request: NextRequest) {
5   const token = request.cookies.get('token')?.value
6
7   // console.log(token, 'this is token')
8
9   if (!token) {
10     return NextResponse.redirect(new URL('/login', request.url))
11   }
12
13   return NextResponse.next()
14 }
15
16 export const config = {
17   matcher: '/dashboard/:path*',
18 }
19
```

Fig:4

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

4.1 CONCLUSION

In conclusion, the internship and the development of the **Far Western University Incubation Center Website** proved to be a highly valuable experience. By working independently on both the frontend and backend, I not only strengthened my technical knowledge in **Next.js, Tailwind CSS, React, Laravel, and MySQL**, but also improved my problem-solving, research, and self-learning skills.

This project gave me practical exposure to the **full software development lifecycle**, from requirement gathering, UI/UX design, development, testing, and deployment. I learned how to build a system that addresses **real-world needs** such as application management, applicant records, and news publishing.

Personally, the internship taught me to be more **independent, disciplined, and confident**, as I managed all parts of the project by myself. Professionally, it showed me the importance of writing clean code, maintaining structured documentation, and ensuring user-friendly designs. The website I created can serve as a **useful platform for the Incubation Center**, helping them manage their programs and students efficiently.

This experience has motivated me to keep learning and exploring more advanced web technologies, preparing me for a career as a professional full-stack developer.

4.1 FUTURE SCOPE

The completion of this project opens several future opportunities for learning and professional growth.

Some possible scopes are:

- **Full-Stack Developer Career** – With experience in both frontend and backend, I can pursue roles in IT companies that require end-to-end development.
- **Advanced Specialization** – Exploring advanced frameworks like **Next.js SSR/SSG, Laravel Livewire, and cloud deployment platforms** to build scalable web applications.
- **Freelancing Opportunities** – As the demand for university and organizational websites increases, I can take freelance projects and build customized platforms for clients
- **Further Enhancement of the FWU Incubation Center Website** – Adding more features such as

email notifications, role-based dashboards, analytics, and cloud hosting for real-time usage.

- **Research and Innovation** – Using this project as a foundation to explore **startup incubation platforms** and contribute to academic and professional research in web technologies.

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